

Small Groups

Arham Lodha

November 02, 2023

Problem M.2a. Prove that the set $\text{Aut } G$ of automorphisms of a group G forms a group, the law of composition being composition of functions.

Proof. 1. **Identity:** id_G is an isomorphism and a function from a group to itself. Thus $\text{id}_G \in \text{Aut}(G)$. Additionally $\forall f \in \text{Aut}(G), \forall g \in G, f(\text{id}_G(g)) = f(g)$ and $\text{id}_G(f(g)) = f(g)$. Thus $f \circ \text{id}_G = \text{id}_G \circ f$.

2. **Inverses:** By definition of an automorphism, $\forall f \in \text{Aut}(G)$ it is an isomorphism and thus has a unique inverse. Furthermore if $f \in \text{Aut}(G)$ then $f^{-1} \in \text{Aut}(G)$.

3. **Closure:** $\forall f_1, f_2 \in \text{Aut}(G)$, by properties of composition $f_1 \circ f_2$ is a function which takes G to G . Furthermore $\forall g, h \in G, f_1(f_2(gh)) = f_1(f_2(g)f_2(h)) = f_1(f_2(g))f_1(f_2(h))$ thus $f_1 \circ f_2$ is a homomorphism. Since f_1, f_2 are both bijections their composition must be as well. Thus $f_1 \circ f_2 \in \text{Aut}(G)$ □

Problem M.2b. Prove that the map $\phi : G \rightarrow \text{Aut}(G)$ defined by $g \rightarrow (\text{conjugation by } g)$ is a homomorphism, and determine its kernel.

Proof. $\forall g, h \in G, \phi(gh) = [a \rightarrow (gh)a(gh)^{-1}] = [a \rightarrow ghah^{-1}g] = [a \rightarrow gag^{-1}] \circ [a \rightarrow hah^{-1}] = \phi(g) \circ \phi(h)$. Thus ϕ is injective. $\forall k \in \ker \phi, \forall g \in G, \phi(k)(g) = kgk^{-1} = g$. Thus $\forall g \in G, kg = gk$, thus $k \in Z(G)$. Thus $\ker \phi \subseteq Z(G)$. $\forall k \in Z(G), \forall g \in G, \phi(k)(g) = kgk^{-1} = gkk^{-1} = g$. Thus $\phi(k) = \text{id}_G$ and $k \in \ker \phi$. Thus $Z(G) \subseteq \ker(\phi) \implies \ker(\phi) = Z(G)$. □

Problem M.2c. The automorphisms that are obtained as conjugation by a group element are called inner automorphisms. Prove that the set of inner automorphisms, the image of ϕ , is a normal subgroup of the group $\text{Aut } G$.

Proof. $\text{Im } \phi$ is a subgroup because image of a homomorphism is a subgroup. $\forall h \in G, f = \phi(h)$ is an inner automorphism. $\forall a \in \text{Aut}(G)$ and $\forall g \in G, (a \circ f \circ a^{-1})(g) = a(f(a^{-1}(g))) = a(ha^{-1}(g)h^{-1}) = a(h)ga(h^{-1}) = a(h)ga(h)^{-1} \in \phi_*(G)$. Thus $\phi_*(G) \trianglelefteq \text{Aut}(G)$ □

Problem M.3a. Determine the group of automorphisms (see Exercise M.2) of the group C_4

Solution. $\forall f \in \text{Aut}(C_4), f(x) = x^k$ where $C_4 = \langle x^k \rangle$ thus $f(x^a) = x^{ka}$. By a previous homework, x^k only generates $C_4 \iff \gcd(k, 4) = 1$, thus $k = 1, 3$.

Define $\phi_m : C_4 \rightarrow C_4$ s.t $\forall g \in C_4 : \phi_m(g) = g^m$. Thus $f = \phi_k$. Thus $\text{Aut}(C_4) = \{\phi_1, \phi_3\}$. Thus $\text{Aut}(C_4) \cong C_2$. ■

Problem M.3b. Determine the group of automorphisms (see Exercise M.2) of the group C_6

Solution. $\forall f \in \text{Aut}(C_6)$, $f(x) = x^k$ where $C_4 = \langle x^k \rangle$ thus $f(x^a) = x^{ka}$. By a previous homework, x^k only generates $C_4 \iff \gcd(k, 6) = 1$, thus $k = 1, 5$.

Define $\phi_m : C_6 \rightarrow C_6$ s.t $\forall g \in C_6 : \phi_m(g) = g^m$. Thus $f = \phi_k$. Thus $\text{Aut}(C_6) = \{\phi_1, \phi_5\}$. Thus $\text{Aut}(C_6) \cong C_2$. ■

Problem M.3c. Determine the group of automorphisms (see Exercise M.2) of the group $C_2 \times C_2$

Proof. $C_2 \times C_2 = \langle x, y \mid x^2 = y^2 = e, xy = yx \rangle$. The elements x, y, xy are all elements with order 2 s.t any multiplication of the two produces the third. Any automorphism will send elements to elements of the same order. Thus any automorphism permutes the 3 elements. Thus $\text{Aut}(C_2 \times C_2) \subseteq S_3$. $\text{Aut}(C_2 \times C_2)$ acts on set $\{x, y, xy\}$. $\exists f \in \text{Aut}(C_2 \times C_2)$ which swaps x and y and fixes xy . $\exists g \in \text{Aut}(C_2 \times C_2)$ which fixes y and swaps x and xy . $\text{Aut}(C_2 \times C_2)^x$ fixes x and freely permutes y and xy . Thus $\text{Aut}(C_2 \times C_2)^x \cong S_2$. Thus $|\text{Aut}(C_2 \times C_2)| = 3 * 2 = 6$. Thus $\text{Aut}(C_2 \times C_2) \cong S_3$. □

Problem M.3d. Determine the group of automorphisms (see Exercise M.2) of the group D_4

Proof. $D_4 = \langle x, y \mid x^4 = y^2 = e, xy = yx^{-1} \rangle$

Let $A = \text{Aut}(D_4)$. A automorphism must send a elements to elements with the same order. Thus $\forall f \in A$, $f(x) = x^k$ where $k = 1, 3$ because $\langle f(x) \rangle = C_4$ and $f(y) = x^m y$ for $m = 0, 1, 2, 3$.

A acts on C_4 . Thus $|A \cdot x| = 2$. A^x is everything which fixes x , automorphisms in this can change $y \rightarrow x^m y$. Thus $|A^x| = 4$. Thus By orbit stabilizer, $|A| = 8$. Define $\alpha \in A \ni \alpha(x) = x^3$ and $\alpha(y) = y$ and $\beta_m \in A \ni \beta_m(y) = x^m y$ and $\beta_n(x) = x$.

$(\alpha \circ \alpha)(x) = \alpha(\alpha(x)) = \alpha(x^3) = x^9 = x$. Since $\alpha \circ \alpha$ fixes x, y (y by definition of α), it fixes D_4 . Thus $\alpha \circ \alpha = \alpha^2 = \text{id}$.

$(\beta_1 \circ \beta_1)(y) = \beta_2(y) = x^2 y$. $(\beta_2 \circ \beta_1)(y) = \beta_3(y) = x^3 y$. Finally $(\beta_3 \circ \beta_1)(y) = x^4 y = y = \beta_0(y) = \text{id}(y)$. Thus $\beta_4 = \text{id}$.

Finally $(\beta_1 \circ \alpha)(x) = \beta_1(x^3) = x^3$ and $(\beta_1 \circ \alpha)(y) = xy$ and $\alpha \circ \beta_1^{-1} = \alpha \circ \beta_3$. $(\alpha \circ \beta_3)(x) = x^3$ and $(\alpha \circ \beta_3)(y) = \alpha(x^3 y) = xy$. Thus $\beta_1 \circ \alpha = \alpha \circ \beta_1^{-1}$.

Thus A follows all relations of D_4 and has the size of D_4 . Thus $A \cong D_4$ □

Problem M.3e. Determine the group of automorphisms (see Exercise M.2) of the group Q_8

Proof. Solved below □

Problem 1. Using our description via generators and relations explain why $\text{Aut}(D_n)$ as a set is equal to $Z/nZ^* \times$ the set of reflections. In particular its order is $|Z/nZ^*|n$ — recall $(Z/nZ)^*$ means the subset of elements of Z/nZ which are invertible (under multiplication. Note this is a finite (multiplicative) Abelian group.

Solution. $\forall A \in \text{Aut}(D_n)$, A must preserve all relations of D_n thus sending elements to other elements of the same order. $A(x) = x^k$ where $\gcd(n, k) = 1$ because $o(x^k) = n \iff \gcd(n, k) = 1$ and $A(y) = x^m y \ni m = 0, \dots, n-1$. $x^m y$ is a reflection in D_n . The map between $Z/nZ^* \times$ the set of reflections $\rightarrow \text{Aut}(D_n)$ is $(k, x^m y) \rightarrow [x \rightarrow x^k, y \rightarrow x^m y]$. The map is injective and surjective thus as sets $\text{Aut}(D_n)$ is equal to $Z/nZ^* \times$ the set of reflections. ■

Problem 2. $\text{Aut}(G)$ of course acts on G . Show $\text{Aut}(D_n)^Y$ is isomorphic to Z/nZ^* (where the notation means the stabilizer of the reflection Y). Is this a normal subgroups of $\text{Aut}(D_n)$?

Proof. $\forall f \in \text{Aut}(D_n)^Y$ $f(x) = x^k \ni \gcd(k, 1)$ and $f(y) = y$. $\phi : Z/nZ^* \rightarrow \text{Aut}(D_n) \ni k \rightarrow [x \rightarrow x^k, y \rightarrow y]$ is a homomorphism. $\forall k \in \ker \phi$, $\phi(k) = \text{id}$. Thus $\phi(k)(x) = x^k = x$ thus $k = \bar{1}$. Thus map is injective. Since f must map elements of the same order to other elements of the same order, $\forall f \in \text{Aut}(D_n)^Y$ $f(x) = x^k \ni \gcd(k, 1)$ and $f(y) = y$. $k \in Z/nZ^*$ by definition, thus $\phi(k) = f$. Thus ϕ is an isomorphism and $\text{Aut}(D_n)^Y \cong Z/nZ^*$

$\forall g \in \text{Aut}(D_n)$, $g(x) = x^m$ and $g(y) = x^k y$ where $m \in Z/nZ^*$, $k \in [0, \dots, n-1]$. $g^{-1}(x) = x^{m^{-1}}$ and $g^{-1}(y) = x^{-km^{-1}} y$. $(g^{-1} \circ g)(y) = g^{-1}(x^k y) = x^{km^{-1}} x^{-km^{-1}} y = y$.

$\forall \sigma \in \text{Aut}(D_n)^Y$, $\sigma(x) = x^a \ni a \in Z/nZ^*$. $g(\sigma(g^{-1})) = g(\sigma(x^{-km^{-1}} y)) = g(\sigma(x^{-km^{-1}}))g(y) = g(x^{-km^{-1}a})x^k y = x^{-ka+k} y$. Thus $g\sigma g^{-1} \notin \text{Aut}(D_n)^Y$. Thus $\text{Aut}(D_n)^Y$ not normal in $\text{Aut}(D_n)$ $\square$

Problem 3. In particular $\text{Aut}(D_4)$ has order 8. What group is it? Note, if we use our usual generators X, Y , then the 8 auts are all given by $X \rightarrow X^{\pm 1}$, and $Y \rightarrow X^i Y$ for some i in $0, \dots, 3$.

Proof. Solved in 3d of Artin Miscellaneous above. $\square$

Problem 4. Show that $Q_8 = \langle I, J | I^4 = J^4 = 1, I^2 = J^2, IJ = J^{-1}I \rangle = \{\pm 1, \pm i, \pm j, \pm k\}$ Using this show that $\text{Aut}(Q_8)$ has order 24. Let S be the 6 elements of order 4. Note they come in natural pairs, each element with its inverse. See that we have a natural map $\text{Aut}(Q_8) \rightarrow S_6 := \text{Perm}(\text{the elements of order 4})$. What group do you want $\text{Aut}(Q_8)$ to be? What do you want these six things to be (where else have you seen S_4 acting on 6 things which are naturally in pairs?) The next problem should help.

Proof. Any automorphism must send elements of the same order to other elements of the same order. Since there are only 1 element of order 2 which is -1 . Any automorphism of Q_8 fixes ± 1 . $\text{Aut}(Q_8)$ acts on $\{\pm i, \pm j, \pm k\}$ all elements of order 4. $\text{Aut}(Q_8)$ can send i to j , j to k , and k to i and i to $-i$. $\text{Aut}(Q_8)$ can send i to any one of the $\{\pm i, \pm j, \pm k\}$. Thus $|\text{Aut}(Q_8) \cdot i| = 6$. Furthermore $\text{Aut}(Q_8)^i$ must fix $\pm i$ but preserves the $\pm j, k$ pairs and thus can only send $j \rightarrow \pm j$ and $k \rightarrow \pm k$. Thus $|\text{Aut}(Q_8)^i| = 4$. Thus $|\text{Aut}(Q_8)| = 24$. We want $\text{Aut}(Q_8) \cong S_4$. The six things should be faces. $\square$

Problem 5. Consider S_6 acting on 6 things, S . divide the elements of S into pairs, and let $i : S \rightarrow S$ be the "involution" which takes each element to its "buddy" (the other element of its pair). Involution just means $i^2 = \text{id}$. Let H be the subgroup of S_6 of elements that are both compatible with the equivalence relation (given by the pairs), and which commute with i . Show that H is isomorphic to S_4 , and that this inclusion of S_4 in S_6 is conjugate to the inclusion of S_4 in S_6 we get by realizing S_4 as the rotational symmetries of a cube, acting on the 6 faces (which we note come in pairs. What is the involution that this S_4 commutes with?).

Solution. Let S be a set with six elements split in 3 pairs, (A pair is a equivalence class). Thus S can be split into a set of pairs, $\{(a, b), (c, d), (e, f)\} = \{P_1, P_2, P_3\}$. Let the involution, i , on the set take an element to the other element in the pair. Let $W \subset S_6$ be the subgroup of S_6 which preserves the equivalence classes. The permutation of the 3 equivalence classes is in W so $\phi : W \rightarrow S_3$ is surjective where $w \in W$ maps to how it permutes the pair. $|S_3| = 6$. $\forall k \in \ker \phi$, k fixes all the pairs and just permutes the elements within each pair. Thus $|\ker \phi| = 2^3 = 8$. Thus $|W| = 48$.

S_4 is the subgroup of S_6 which corresponds to the rotational symmetries of the cube. It also acts on 6 things, the faces of a cube and preserves opposites. Thus there is a completely natural equivalence class on the faces of a cube where 2 faces are equivalent iff they are opposites. Thus the set of 6 faces gets split up into pairs. Thus $S_4 \subset W$. However S_4 also commutes with $-I \in \text{Rig}(R^3)$, an involution which swaps opposites. $[W : S_4] = 2$.

Thus subgroup H of W is the subgroup which commutes with the involution i on the set X . H acts on S . $|H \cdot a| = 6$ because a can go to any element in the 6 set. $\forall f \in H^a$ since f must fix the pairs $f(b) = b$ so all f can do is permute P_2 and P_3 and/or swap elements within each pair thus $|H^a| = 4$. Thus $|H| = 24$ and $[W : H] = 2$ and $H \cong S_4$. Thus $\text{Aut}(Q_8) \cong S_4$.

By the previous homework problem 3, we know the we know that the inclusions are conjugate. ■